



Screening green solvents for multilayer plastic film recycling processes

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ABSTRACT

Multilayer (ML) plastic films are essential packaging materials that help protect products from diverse external factors; however, only 5% of all ML films are recycled in the United States. Solvent-based technologies are a promising alternative for recycling ML films because they enable recovery of constituent polymer resins. For example, the Solvent Targeted Recovery and Precipitation (STRAPTM) process sequentially dissolves and separates polymer components using a series of targeted solvent washes. A crucial design aspect of this process is the impact of selected solvents on human health and on the environment. This work introduces a computational framework that integrates molecular modeling, process modeling, techno-economic analysis (TEA), and life-cycle analysis (LCA) to quickly screen green solvents for solvent-based ML recycling processes. Initial screening for solvents based on selectivity is performed by estimating temperature-dependent solubilities using molecular-scale models. Subsequent screening uses basic estimates of energy use and octanol-water partition coefficients (logP) as key measures of health, safety, and environmental hazards. Detailed process modeling, TEA, and LCA are used on a reduced set of promising solvents identified in early screening steps to more accurately determine how solvent selection and associated operating conditions impact overall economics and environmental impacts. The framework is used for the identification of green solvents (from a database of 1,000 solvents) that separate an industrial ML film composed of polyethylene (PE), ethylene vinyl alcohol (EVOH), and polyethylene terephthalate (PET). Our analysis shows the effectiveness of the framework and reveals fundamental trade-offs between solvent greenness, solubility, and economics. Our work emphasizes the importance of taking a holistic systems view during solvent design and aims to inform the development of new processes for ML film recycling and the identification of new ML films that are easier to recycle.

1. Introduction

Multilayer (ML) plastic films have been pivotal in promoting product safety, security, and handling (Alias et al., 2022; Anukiruthika et al., 2020; Ashok et al., 2016; Matthews et al., 2021). However, their predominant disposal in landfills or the environment at their end-of-life raises significant environmental concerns (Bauer et al., 2021; Horodytska et al., 2018; Sánchez-Rivera et al., 2023; Zhao et al., 2018). ML films can be composed of different polymers (e.g., PE, EVOH, PET) and up to hundreds of layers that serve diverse functions in prolonging the shelf life of products (Morris, 2022; Niaounakis, 2019). Unfortunately, the intricate combination of polymer layers complicates the recycling process; only 5% of ML plastic films are currently recycled in the United States (the rest are diverted to incineration facilities or landfills) (Hutchings et al., 2021; Milbrandt et al., 2022).

Several technologies are currently being explored for recycling plastic waste (Li et al., 2022), including thermochemical processes such as pyrolysis (Li et al., 2023; Wang et al., 2023) and thermal oxo-degradation (Olafasakin et al., 2023), physicochemical processes such as supercritical fluid separation (Anouar et al., 2015), and mechanical processes (Jeswani et al., 2021; Suzuki et al., 2022). The selection of a recycling technology significantly hinges on the intended final product, which influences process design, economics, safety, and environmental impact. For instance, pyrolysis of plastic waste primarily targets the production of pyrolysis oil, which can be processed into high-value chemicals, thus altering the original polymer structure. Mechanical recycling preserves the polymer structure and is scalable but cannot be used for complex materials that involve multiple polymers, such as ML films (Lisiecki et al., 2024; Sánchez-Rivera et al., 2023).

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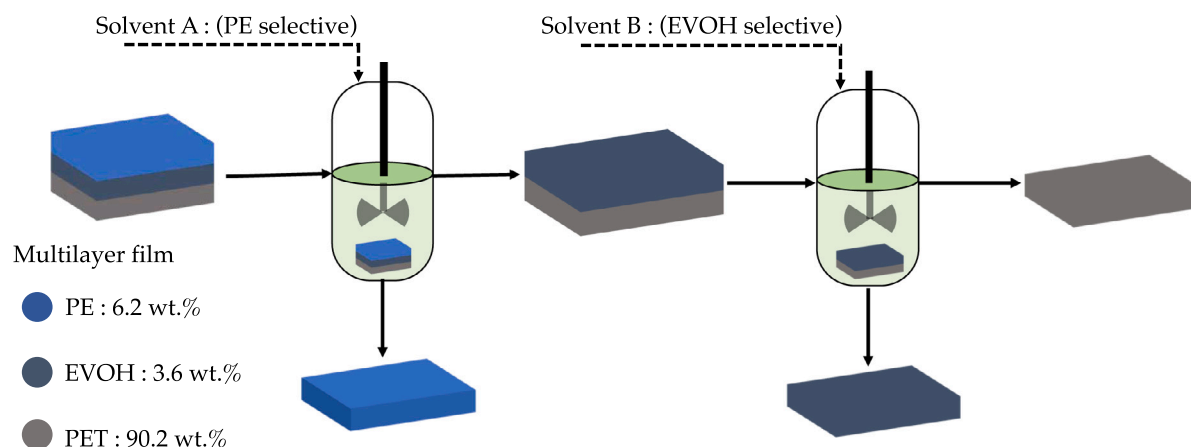


Fig. 1. Illustration of multilayer plastic film separation using STRAP™. This recycling process uses selective solvents for separating individual polymer layers. The solvents need to be carefully selected to enable high and selective solubility of target polymers at reasonable temperatures (as dissolution temperature impacts energy usage). After solubilizing a target polymer layer, the solvent-polymer mixture is cooled down to enable polymer precipitation and recovery (this cooling step also impacts energy usage).

Solvent-based separation technologies have recently emerged as an alternative for recovering resins with near-virgin properties from ML plastic films (Zhao et al., 2018). An illustrative embodiment of this methodology is the Solvent-Targeted Recovery and Precipitation (STRAP™) process introduced by Walker et al. (2020), see Fig. 1. This approach systematically separates the polymer constituents of ML plastic films through dissolution and precipitation by employing selective solvents and by exploiting temperature-dependent solubility of polymers in solvents (Sánchez-Rivera et al., 2021). A critical step in designing this process is solvent selection/design, which serves as the linchpin of the entire approach and which requires the prediction of temperature-dependent polymer solubilities in many possible solvents. To address this need, molecular-scale models have been used to predict the solubilities of various plastic components, including but not limited to PE, EVOH, and PET, in over 1000 solvents (Zhou et al., 2023; Munguía-López et al., 2024). These predictions are made by modeling polymers as short oligomers and then employing molecular dynamics simulations and the COnductor-like Screening MOdel for Real Solvents (COSMO-RS) to predict thermodynamic quantities that dictate solubility.

Solvent-based separation technologies have demonstrated their efficacy in handling diverse plastic waste feedstocks such as face masks (Yu et al., 2023) and reverse-printed films (Sánchez-Rivera et al., 2023); however, environmental and health concerns associated with using solvents to upcycle certain materials (e.g., food-contact materials) place limitations on such technologies. In addition, when addressing ML plastic recycling challenges by deploying solvent-based approaches, it is imperative to mitigate the emergence of new toxic wastestreams from harmful solvents to avoid burden-shifting (Algunaibet and Guillén-Gosálbez, 2019). Although high solvent recycling rates are ensured in the solvent-based recycling process, small quantities of these solvents potentially escape into the environment or are deliberately disposed of after several uses. The ecological impact of such waste streams can be substantial when the solvents in use possess toxic attributes. Moreover, using toxic solvents may pose risks to the safety of operation personnel. Similarly, the choice of solvent impacts the process economics as the use of different solvents will impact energy requirements (e.g., due to changes in temperature-dependent solubilities). Therefore, it is crucial to incorporate toxicity considerations in selecting solvents and understand their influence on process design, economics, and environmental impacts in a holistic manner (Hessel et al., 2022; Winterton, 2021).

Previous work has sought to define what a “green solvent” is by using green chemistry principles. Saleh and Koller (2018) characterize green solvents as those possessing low reactivity and flammability, exemplified by ionic liquids. Choi et al. (2011) initially identified

natural deep eutectic solvents (NADES) and deemed them green due to their derivation from plant sources. A review by Janicka et al. (2022) classifies supercritical fluids as green solvents, as they can dissolve materials at temperatures and pressures exceeding their critical points without necessitating purification during recovery or reuse. Knez et al. (2014) examined industrial applications of supercritical fluids and delineated the associated risks of high-pressure processes and elevated operational costs.

Considering these perspectives, the greenness of solvents can be broadly quantified using the following criteria: (1) whether they are derived from a renewable source, (2) their usage efficiency, (3) health, safety, and environmental hazards, (4) recyclability, and (5) biodegradability. The ideal green solvent, designed to harmonize sustainability and technical efficiency, should ideally originate from a renewable source (e.g., plant-derived solvents) and maintain cost-effectiveness as a fundamental criterion. It should also have benign health, safety, and environmental characteristics throughout its life-cycle. In addition, it should be easily recyclable or biodegradable after disposal. It is important to highlight that the utility of the green solvent in the actual process is of particular importance (e.g., high solubility and reduced energy consumption), as this impacts the overall process efficiency and greenness of the entire process. Although it is desirable to find an ideal solvent that satisfies all these greenness criteria, it is difficult (if not impossible) to simultaneously achieve all these goals simultaneously.

Several studies have sought to identify and design green solvents, but, to the best of our knowledge, no existing study has developed a holistic framework to evaluate and select green solvents within the context of a solvent-based plastic recycling process (Ikegwu et al., 2024). Therefore, in this study, we aim to develop a green solvent screening framework for separating ML plastic films into their polymer constituents. This computational framework is holistic and considers polymer solubility, various solvent greenness metrics, and the impact of solvent selection on the overall process design and economics. Although this framework was developed in the context of ML plastic film separation, it can be adapted for any solvent-based process. This work can also guide the design of new green solvents with relevant properties for ML plastic film separation and inform the design of easy-to-recycle ML plastic films.

The paper is structured as follows: Section 2 details the computational framework and its various elements. Section 3 present an application of the framework to a practical case study involving the separation of an industrial ML plastic film, illustrating the utility of the framework and highlighting how green solvent selection informs process design. Finally, Section 4 offers our concluding remarks on this study and identifies potential avenues for further studies.

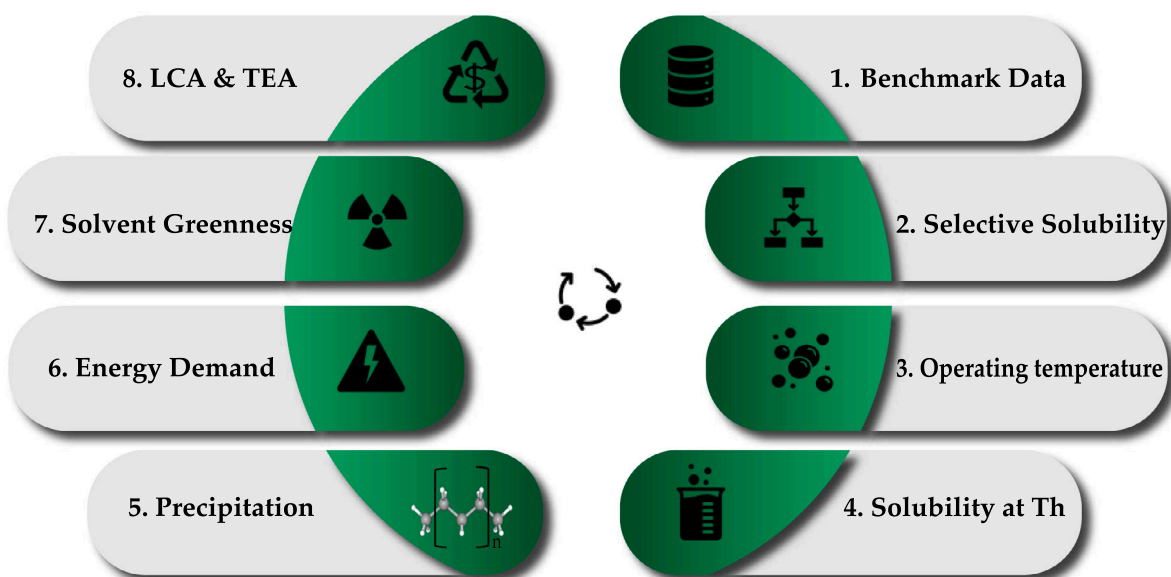


Fig. 2. Green solvent selection framework for ML film separation. (1) Benchmark solvent solubility data at 120 °C. (2) Choose solvents that selectively dissolve the target polymer. (3) Choose solvents with boiling points above room temperature (RT) and estimate the highest operating temperature (Th) at 1 atm. (4) Select solvents with at least 15 wt.% solubility of the target polymer at their respective highest operating temperature (Th). (5) Choose solvents with low solubility of the target polymer at RT. (6) Evaluate the dissolution energy requirement for each solvent. (7) Assess the environmental impact and governing regulations of the selected solvents. (8) Identify trade-offs between environmental metrics and process economics via life cycle assessment (LCA) and techno-economic analysis (TEA).

2. Computational framework

The steps/elements of the proposed framework that screens for green solvents in the context of efficiently separating ML plastic films into their constituent virgin resins is sketched in Fig. 2. We highlight that these steps can be highly computationally intensive due to the large number of solvent-polymer combinations that need to be considered; as such, our goal is to enable fast screening/filtering of solvents in early steps using simple computational procedures and metrics and increase the level of detail and sophistication in later steps.

2.1. Polymer solubility prediction and selective dissolution

The framework begins by predicting polymer solubilities in over 1000 solvents at a standard temperature of 120 °C (Step 1). This temperature is chosen to prevent thermal degradation of the polymers (Pielichowski et al., 2022) while providing adequate temperature leeway to achieve desired solubilities, as discussed by Zhou and coworkers (Zhou et al., 2023). The target polymers were modeled as short oligomers in reference solvents using molecular dynamics simulations to obtain different conformations from corresponding simulation trajectories. These conformations are then optimized and serve as inputs into CONductor-like Screening MOdel for Real-Solvents (COSMO-RS) (Zhou et al., 2021) to predict temperature-dependent solubilities at the infinite dilution limit. This molecular simulation approach has been previously shown to predict temperature-dependent polymer solubilities in reasonable agreement with experimental measurements (Zhou et al., 2023).

To ensure solvent selectivity in Step 2, the solubility difference between the target polymer and other polymers in the ML films should be greater than 10 wt.% (Zhou et al., 2023). In addition, the solvents with boiling points lower than room temperature (RT) are removed in Step 3; this rule only applies to processes that wish to operate at standard atmospheric pressure and is intended to avoid high solvent volatilization and extremely low operating temperatures. Similarly, the highest operating temperature (Th) is set to 1 °C less than the solvent boiling point in Step 3 to maintain this operating condition. Solvents with at least 15 wt.% solubility for the target polymer at Th are then

selected in Step 4. This solubility range has been identified to be reasonable for ML plastic film separation (Sánchez-Rivera et al., 2023). To recover the dissolved layer after filtering the solution to remove residual solids (i.e., undissolved polymers) at the high operating temperature, solvents with a low solubility of the target polymer at room temperature are selected in Step 5 to ensure easy precipitation after cooling the polymer solution (Walker et al., 2020). Although Steps 2–5 pertain specifically to selective procedures conducted at standard atmospheric pressure, other aspects of this framework are generalizable to solvent-based processes.

2.2. Energy demands and screening

In Step 6, we assess the environmental impact of the selected solvents from a process perspective by using the minimum energy required by each solvent. This is done by using a simple energy balance $Q = mC_p\Delta T$. Here, Q is the total energy intensity (in J), m (in grams) is the mass of the solvent, C_p (in $\text{J g}^{-1} \text{K}^{-1}$) is the heat capacity of the solvent assuming temperature-independence, and ΔT (in K) is the temperature difference between RT and Th (this difference capture the operating range of the solvent in the process). The measure Q is simple but comprehensive; this captures the overall carbon intensity originating from employing a solvent within the process, as it captures the total energy required to adjust the solvent temperature across the entire operating range in the process. Upon calculation of Q , the solvents are ranked using these estimated minimum energy requirements from lowest to highest.

2.3. Octanol-water partition coefficient (logP) prediction

The octanol-water partition coefficient (logP) is a fundamental molecular property that provides information about the tendency of a solvent to bioaccumulate in living organisms. Several regulations around the world, including the United States, have emphasized estimating the solvent logP prior to their usage (Renner, 2002). This quantity is often described as the equilibrium ratio of concentrations of a chemical in a mixture of octanol (hydrophobic or non-polar phase) and water (hydrophilic or polar phase) (Cevc, 2015). Several

empirical and physics models have been developed to predict compound partition coefficients, including quantitative structure-retention relationship (QSRR) models (Qiao et al., 2023), solvation models based on electronic density (SMD) and COSMO-RS methods (Van der Spoel et al., 2019), and physics-informed machine learning models (Zhu et al., 2022). Van der Spoel et al. (2019) showed that the COSMO-RS model performs well compared to other models and values from the PubChem public database.

In **Step 7** of our framework, we use the COSMO-RS model to calculate logP values of all solvents, where logP for solvent i is computed using Eq. (1).

$$\log_{10}(P_i) = (\mu_i^{(1)} - \mu_i^{(2)}) / RT \ln(10) + \log_{10}(V_q) \quad (1)$$

We use the notation logP for solvent i as shorthand for $\log_{10}(P_i)$. The chemical potentials $\mu_i^{(1)}$ and $\mu_i^{(2)}$ of all solvents i in infinite dilution in pure octanol (phase 1) and water (phase 2) are automatically computed, respectively. The quantity V_q is the volume quotient between octanol and water, estimated from their densities. Solvents with logP > 0 are considered lipophilic and thus have higher bioaccumulation tendencies; solvents with logP < 0 are considered hydrophilic, indicating lower bioaccumulation tendencies. Although solvents with logP > 0 are generally not regarded as green, there are no clear indications on what the acceptable logP threshold is for food packaging applications. Hence, we consider only solvents with logP ≤ 3 as an initial screening process in Step 7 of our framework. We use this to assess the trade-offs among the different selected solvents.

2.4. Solvent toxicity database and regulatory framework

Solvent toxicity is a crucial aspect to consider in the design and operation of any chemical process (Diorazio et al., 2016). This has inspired the development of quantitative toxicity metrics for common solvents that are used for guiding solvent selection. These guides mostly involve environmental implications of each solvent usage. Some of these include industrial guidelines developed by GlaxoSmithKline (GSK) (Henderson et al., 2011), the American Chemical Society (ACS) Green Chemistry Institute Pharmaceutical Roundtable (ACS GCI-PR) (Prat et al., 2014), and AstraZeneca (AZ) (Diorazio et al., 2016).

The GSK guideline categorized 111 solvents with over 10 primary functional groups under 6 headings. These categories include waste, environmental impact, health, flammability, reactivity, and life cycle. Scores between 1 and 10 were assigned to each solvent on the GSK guideline, with 1 indicating the solvents with the most concern and 10 indicating the solvents with the least concern. A couple of extra categories were included to classify solvents based on legislation and restriction bans placed by the Environmental Health Services (EHS). Conversely, AZ and ACS GCI-PR guidelines categorized 46 and 63 solvents, respectively, with a score of 10 indicating the solvent with the most concern and 1 with the least concern. Although these guidelines contain different metric categories, they all seek to rate solvents based on the Health, Safety, and Environment (HSE) metrics.

This study employs only the GSK solvent guideline. This guideline contains the most solvents, covers the HSE metrics, and includes the legislation and regulatory ban information about some solvents. Although the GSK guideline can be used with other guidelines, the difference in the meaning of the rank assigned to each solvent makes it more difficult. Moreover, using these guides is not the sole greenness check in this framework, as these guidelines do not contain information about most of the solvents in our database. Hence, other solvent regulatory standards such as the Registration, Evaluation, Authorization, and Restriction of Chemical Regulation (REACH) initiative and the Toxic Release Inventory (TRI) were adopted to discard solvents of concern. At the time of this study, benzene, toluene, formamide, and fluorinated solvents were included on the list of restricted substances (<https://echa.europa.eu/substances-restricted-under-reach>, Byrne et al., 2016, <https://www.epa.gov/toxics-release-inventory-tri-program/tri-listed-chemicals>). These solvents were subsequently removed from our dataset in Step 7.

2.5. Process simulation and techno-economic analysis (TEA)

The overall recycling process cost is an important economic factor when selecting solvents. While the cost of a solvent directly impacts total process cost, other process costs associated with using a specific solvent (such as energy costs affected by operating conditions) need to be accounted for. Hence, it is necessary to simulate the recycling process, conduct detailed material and energy balances, and carry out a detailed TEA that provides a more holistic cost evaluation. Our analysis captures total cost using the minimum selling price (MSP) of the recovered polymer resin; this is the price needed to reach a net present value of zero for the process over a specified period. The process simulation and TEA (**Step 8** of our framework) are performed using BioSTEAM, an open-source Python process simulator that has been benchmarked against proprietary software such as Aspen Plus (Cortes-Pena et al., 2020).

2.6. Life cycle assessment (LCA)

LCA is a well-developed methodology for estimating the environmental impact of processes such as solvent-based plastic recycling (del Carmen Munguía-López et al., 2023). In addition to solvent screening using the solvent toxicity database and regulatory framework earlier mentioned, an LCA is performed to estimate the impacts of using each solvent within the process. These impacts include the human toxicity impact in cancer/non-cancer Comparative Toxic Units for humans (CTUh), the freshwater ecotoxicity in Comparative Toxic Units for aquatic ecotoxicity (CTUe), and the climate change impact in kg CO₂ eq. units. OpenLCA and BioSTEAM are used to perform this assessment in **Step 8** of our framework. The impact of producing 1 kg of virgin resins is estimated using Eq. (2).

$$I_i = \sum_u IF_{i,u} \times \dot{m}_u \quad (2)$$

where i denotes the impact categories and u represents the utilities (electricity (MJ), steam (MJ), and cooling water (m³)). Therefore, the impact factor of a category (I_i) is the sum of the product of the impact factor of producing each utility ($IF_{i,u}$) and the flow rate of each utility ($\dot{m}_u$) required to produce 1 kg of virgin resins via the STRAP™ process. The environmental footprint (mid-point indicator) method from the OpenLCA 2.0 environmental footprint database is employed for this study.

It is important to highlight that conducting detailed process modeling, TEA, and LCA is time-consuming and should thus be conducted only on a selected number of solvents. This is why it is critical to develop fast screening tools in earlier steps of the framework.

3. Case studies

We demonstrate the use of our framework using a case study in which we use STRAP™ for separating an industrial ML film with 3 polymer components, see Fig. 1. The ML film is produced by the Amcor company and consists of 6.2 wt.% PE, 3.6 wt.% EVOH, and 90.2 wt.% PET (del Carmen Munguía-López et al., 2023). Our study delineates a couple of distinct scenarios originating from a systematic, bottom-up approach. The first scenario focuses on the identification of a green solvent to separate a 2-component film (EVOH-PET), which corresponds to selecting Solvent B for the process depicted in Fig. 1. The second scenario pertains to the identification of green solvents for separating PE from a three-component film (PE-EVOH-PET), which corresponds to selecting Solvent A for the process depicted in Fig. 1.

As depicted in Fig. 2, Step 1 of the framework involves screening for solvents that meet the solubility criteria for recovering all polymers within the ML plastic film. By adopting the solubility prediction framework introduced by Zhou et al. (2023), the solubilities of PE, EVOH, and PET in 1000 solvents were predicted at different temperatures

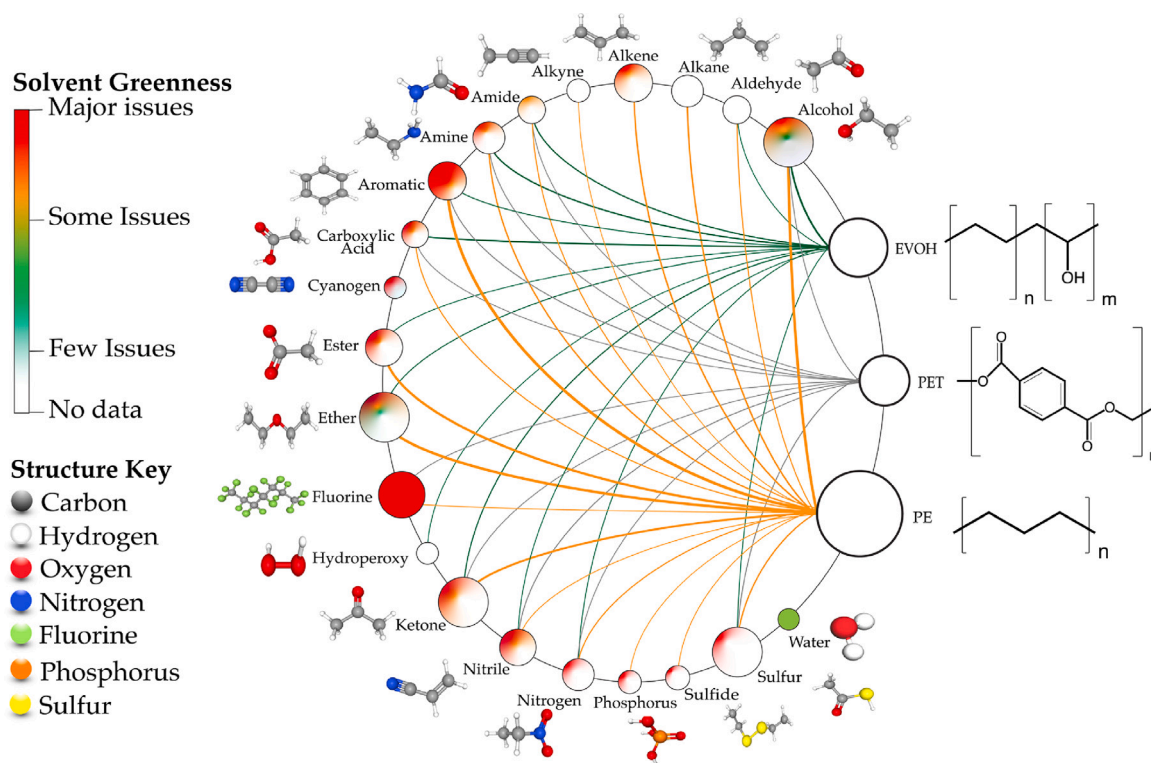


Fig. 3. Visualization of the toxicity and solubility database at the operating temperature (T_h) for each solvent. Nodes on the left represent solvent classes, with node size indicating the number of solvents per class. Nodes on the right represent the three target polymers, with node size indicating the number of solvents capable of dissolving them. Edge sizes reflect the number of solvents able to dissolve the connecting polymer. Color gradients in solvent nodes indicate the greenness level of solvents within each class based on the GSK solvent selection guidelines.

ranging from 30 °C to 120 °C to obtain a solubility database. Fig. 3 provides a visualization of the solubility predictions in this set of solvents for these 3 polymers at each solvent's respective high operating temperature (T_h), which is either a degree less than its boiling point or a maximum of 120 °C. The set of solvents are split into distinct classes based on their functional groups, to convey the breadth and diversity of the solvent space considered in this study. The illustration also shows the difficulty in finding solvents that can dissolve PET, as indicated by the small size of its node and the number of connecting edges. This difficulty compounds when trying to selectively dissolve PET while leaving the other two polymers undissolved because no solvent class node connects only to PET. The visualization shows that PE has more solvents that can be used to selectively dissolve it. For example, solvents with alkane, alkene, and alkyl groups preferentially dissolve PE. The color gradient within the solvent class nodes reflects the greenness ratings of each solvent within a class, according to GSK solvent selection guidelines (Henderson et al., 2011); green indicates the least hazardous concerns and red indicating the most hazardous concerns. While it is observed that red dominates the set of solvents under study (e.g., indicating that organic solvents are not the preferred choice of solvent), there is a need to examine the trade-offs due to the potential process efficiencies offered by organic solvents (e.g., high solubility and less energy usage) (Jimenez-Gonzalez, 2019). In addition, there is a need to more rigorously quantify the greenness of solvents, beyond what is recommended by the GSK guideline.

3.1. EVOH-PET separation

In alignment with the separation sequence outlined by Sánchez-Rivera et al. (2021), our objective is to identify green solvents suitable for the selective dissolution of EVOH from an EVOH-PET polymer blend. A flowchart showing how solvents are screened out at each step of our framework for this case study is shown in Fig. 4. Although these

steps are not necessarily hierarchical, this sequence was adopted for simplification. To ensure the selective dissolution of EVOH, solvents exhibiting a minimum of 10 wt.% difference in solubility between EVOH and PET at 120 °C were retained. This initial screening resulted in the identification of 116 potential solvents from a the entire set of 1000 solvents available. Subsequently, only solvents amenable to operation at ambient pressure (i.e., those possessing boiling points above room temperature), were selected. This criterion led to the further exclusion of 19 solvents, resulting in a remaining set of 97 solvents. The high temperatures (T_h) for each of these solvents were further determined, and solvents exhibiting a minimum 15 wt.% solubility at T_h were selected to ensure the solubility of EVOH. This refinement further reduced the solvent list to 56 candidates. To facilitate subsequent precipitation processes, our selection was refined by retaining only those solvents with less than 2 wt.% solubility of EVOH at room temperature (RT). Consequently, the number of viable solvents was reduced to 45.

The next step in the framework requires assessing solvent greenness. As high energy requirements equate to high carbon emissions, the minimum energy requirements for each of the 45 selected solvents were estimated using the simple energy balance. This calculation only accounts for the minimum energy required to raise the temperature of each solvent to the specified dissolution temperature, which is a function of the specific heat capacity of the solvent and T_h . Although the energy required is a function of the required solvent mass which is a function of solvent-polymer solubility, the energy per mass (J/g) was calculated instead, because the solvent-to-plastic film feed ratio is typically fixed in the STRAP™ process. The temperature-dependent specific heat capacities for the solvents were estimated using the BioSTEAM simulator (which uses a QSPR model). Next, logP of each solvent was estimated and screened to remove solvents with logP greater than 3, although it was found that none of the 45 solvents exceeded this threshold.

The estimated solvent energy requirements, logP, and EVOH polymer solubility are presented in Fig. 5. One of the highlighted data

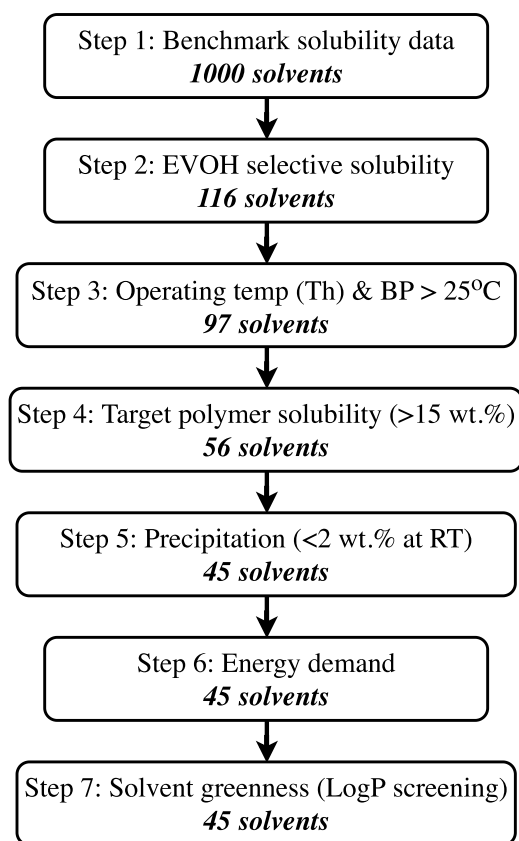


Fig. 4. Green solvent selection flowchart for case-study one (EVOH-PET). This depicts how solvents are sequentially screened out when selecting green solvents to selectively dissolve and precipitate EVOH from an EVOH-PET plastic. Following the framework depicted in Fig. 2, this approach reduces the set of solvents from 1000 to 45. Note that no solvents are eliminated when considering the criteria in Steps 6 and 7.

points (i.e., solvent) indicated by an arrow in Fig. 5 is dimethylsulfoxide (DMSO) with an EVOH solubility of 35 wt.% and PET solubility of 8 wt.%, fulfilling the selective solubility criteria for this separation process. This result agrees with the observations made by Walker et al. (2020) and Zhou et al. (2023), without necessarily considering its greenness. However, as depicted in Fig. 5, there are other factors to consider and a need for trade-off analysis. Some other solvents are also highlighted in the point cloud as shown in Fig. 5. To select a green solvent, we aim to locate solvents at the bottom left corner of Fig. 5, which is indicative of their low energy requirements and negative logP values. Moreover, choosing solvents with high solubility, while beneficial for selective dissolution, poses challenges in recovering food-grade ML plastic films due to stringent solvent retention concentration requirements (Heckman, 2005). This elevated solubility can complicate or increase the energy demands for producing resins containing very low solvent concentrations embedded in the polymer matrix. For example, formic acid, with a lower but sufficient EVOH solubility of 19 wt.%, lower energy requirement of 120 J/g, and a negative logP of -0.6 , could be a better alternative to DMSO as seen in Fig. 5. However, the corrosive nature of formic acid makes it less desirable for recovering food contact materials. Therefore, this begs the need for further qualitative solvent greenness assessment.

The GSK solvent guidelines, the PubChem public database, the REACH initiative, and the Toxic Release Inventory (TRI) databases were used to evaluate the selected solvents qualitatively. Fig. 6 presents a heat map depicting some important solvent greenness metrics for 40 solvents for the selective dissolution and precipitation of EVOH. The full solvent list, labels, and properties are reported in the *Extended*

Supplementary Information (ESI). As seen in Fig. 6, using some solvents may be restricted or not depending on their associated hazards. For instance, solvents like ethylene glycol monomethyl ether, N-methyl formamide, dimethyl formamide (DMF), N,N-dimethyl acetamide, and N-methyl-2-pyrrolidone (NMP), all have restrictions placed on their usage (Henderson et al., 2011).

When selecting a green solvent, the boiling point is crucial. Solvents with high boiling points require significant energy for separation via distillation, especially when dealing with solvent mixtures (del Carmen Munguía-López et al., 2023). Conversely, solvents with low boiling points should generally be avoided due to their volatility at ambient temperatures, posing health risks and requiring substantial energy for condensation. For instance, formic acid, with a boiling point of 101 °C, considerably away from ambient temperature, would be a preferred solvent. In contrast, N-(2-hydroxyethyl)-2-pyrrolidone with a high boiling point of 295 °C is less desirable, as indicated by the red color on the heat map. Furthermore, as previously mentioned, solvents with high logP are considered toxic. For instance, 4-ethyl phenol has a high logP of 2.3, making it not desirable, as marked red in Fig. 6.

In conclusion, a solvent may appear green based on one metric but may not satisfy other green criteria. Also, it is critical to note that the choice of which greenness metric to prioritize depends on the specific application of the solvent. Solvents like acetic acid, DMSO, epsilon-caprolactam, and bromoacetic acid are some recommended solvents for EVOH selective dissolution. Acetic acid appears to be a promising solvent for the selective dissolution of EVOH because it has the lowest logP value and a good LCA metric score of 8 out of 10 based on the GSK solvent screening guide. DMSO also appears to be a promising solvent for this process due to its negative logP and high solubility. Epsilon-caprolactam is another promising solvent for this process due to its negative logP and because it can be derived from biomass sources (Buntara et al., 2011). Lastly, based on the energy requirements, bromoacetic acid appears to be a good solvent.

3.2. PE-EVOH-PET separation

In the previous section, we applied our green solvent selection framework to the selective dissolution of EVOH from an EVOH-PET polymer blend. In this scenario, we will consider screening for green solvents to selectively dissolve PE from a PE-EVOH-PET polymer blend, as this is a common selection for typical ML plastic films (del Carmen Munguía-López et al., 2023; Sánchez-Rivera et al., 2023). Selectively dissolving PE instead of EVOH will lead to a different set of possible solvents due to their interactions with PE and because of the need to avoid dissolving EVOH and PET.

After the first screening step, 576 solvents were found to selectively dissolve PE at 120 °C. This large number of solvents is due to the fact that most organic solvents in the database are nonpolar, as is PE, and results from the heuristic principle of like-dissolves-like. Subsequently, only 456 solvents were amenable to operation at ambient pressure. The solubility of PE at the selected solvent high temperatures (Th) was determined, and 263 solvents exhibited a minimum of 15 wt.% solubility. Finally, to facilitate precipitation at room temperature, we retained only those solvents with less than 2 wt.% solubility of PE at room temperature (RT). This screening step further reduced the candidate solvents to 151.

Using the simple energy balance equation, the energy requirement of each solvent was estimated. The logP for each of the selected solvents was also screened, although all 151 solvents had logP values greater than 0. Fig. 7 shows the predicted energy requirement, PE polymer solubility, and logP for the 151 selected solvents. One of the highlighted data points (solvent) indicated by an arrow in Fig. 7 is toluene, because it was employed by Walker et al. (2020) for a similar separation process. Toluene was previously chosen because it has a predicted PE solubility of 20 wt.%, EVOH solubility of 0 wt.%, and PET solubility of 2 wt.%, which fulfill the selective solubility criteria for this separation

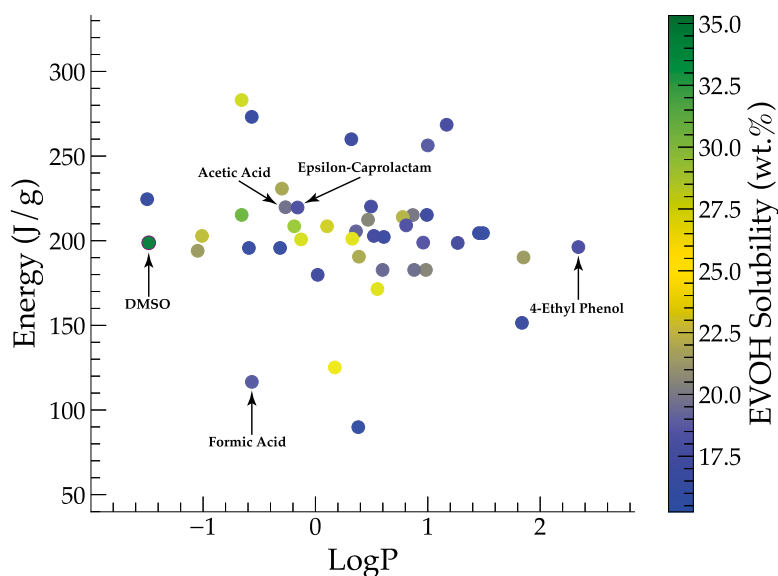


Fig. 5. Predicted energy required, logP, and EVOH polymer solubility of 45 screened solvents. Some solvents in the point cloud are highlighted by arrows showing trade-offs across the solvent space.

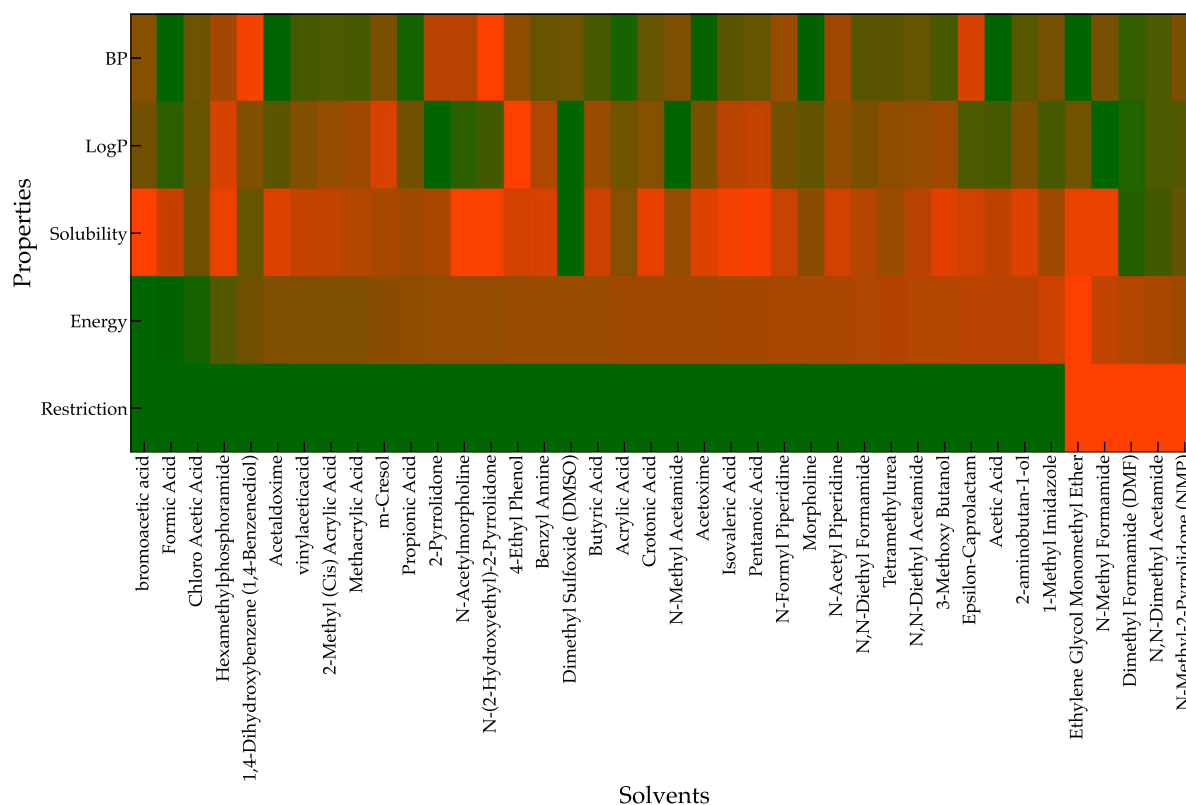


Fig. 6. Selected solvents for EVOH selective dissolution. The green color represents low boiling point, low logP, high solubility, low energy, and no restriction. BP denotes boiling point. The properties of each solvent are presented in the Extended Supplementary Information (ESI).

process. However, its logP value (+2.5) is relatively high, making it a less desirable solvent for recovering food-grade ML plastic film. As seen in Fig. 7, all the selected solvents have positive logP, indicating a high bioaccumulation tendency. This observation highlights the difficulty in efficiently recovering PE from an ML plastic film using organic solvents while considering solvent greenness. Some other solvents are also highlighted in the point cloud as shown in Fig. 7. As previously noted, solvents at the bottom left are more desirable. For instance, this framework, as shown in Fig. 7 suggests that a solvent such as

cyclopentanone with a logP value of 0.5, good solubility of 30 wt.%, and energy requirement of 200 J/g, could be a more desirable solvent for recovering PE compared to toluene.

Furthermore, chlorinated, brominated, and fluorinated solvents were identified as having the lowest energy requirements. However, these solvents present significant environmental and health challenges. This illustrates the holistic nature of the framework in selecting green solvents for ML plastic film separation. These selected solvents were

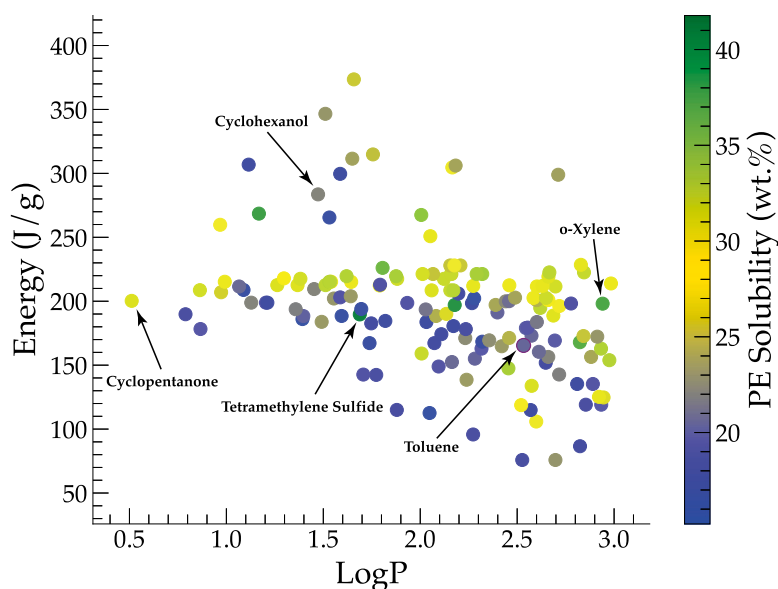


Fig. 7. Predicted energy required, logP, and PE polymer solubility of 151 screened solvents. Some solvents in the point cloud are highlighted by arrows showing trade-offs across the solvent space.

further screened using the solvent selection guidelines and the government regulatory guidelines mentioned earlier to identify solvents with less environmental concerns and regulations. Fig. 8 presents a summary of some important solvent greenness metrics for 40 solvents for the selective dissolution of PE from a PE-EVOH-PET polymer blend. The full solvent list, labels, and properties are reported in the ESI. For example, dichlorobenzene and trichlorobenzene are on the restricted substances list under the REACH initiative (<https://echa.europa.eu/substances-restricted-under-reach>). Similarly, chlorofluorobenzene, and fluoroanisole both have regulatory bans on their use (Henderson et al., 2011). Although toluene has been previously employed for ML plastic film separation (del Carmen Munguía-López et al., 2023; Walker et al., 2020; Sánchez-Rivera et al., 2021), it poses a serious danger to the environment and is on the list of restricted substances under the REACH initiative (<https://echa.europa.eu/substances-restricted-under-reach>).

It is also crucial to screen solvents based on their application. In this case, these solvents are applied to ML plastic films used in the food packaging industry. Hence, it is important to remove solvents that pose any form of health hazard (Mehrabani et al., 2022). For example, public solvent databases like PubChem show that toluene, naphthalene, MIBK, and indene all pose serious health hazards. The ether, ester, and ketone solvent classes were predominantly found for PE selective dissolution upon applying this green solvent selection framework, aligning with the GSK solvent guidelines, which categorize these classes as the greenest solvents (Henderson et al., 2011). Summarily, solvents like tetramethylene sulfide, cyclopentanone, o-xylene, and α -chloro methyl acrylate are some recommended solvents for PE selective dissolution. Tetramethylene sulfide is a good candidate solvent for this separation process due to its high solubility for PE and relatively low energy requirements and logP values. As indicated earlier, cyclopentanone also proves to be a good solvent for recovering PE from this ML plastic film. Other solvents such as o-xylene and α -chloro methyl acrylate also appear to be good solvents because they can be derived from biomass and have low energy requirements.

Our results highlight trade-offs between the different solvent greenness metrics across different solvents. To further analyze these trade-offs, selected solvent pairs and their rationale for selection (see Table 1) are used to compare process cost and environmental impacts via TEA and LCA.

3.3. Process modeling and techno-economic analysis

Following the framework developed in this study, we model the entire separation process (Fig. 1) using the selected solvent pairs presented in Table 1 and estimate the MSP of the recovered polymer resins via techno-economic analyses. A similar process flow diagram, as well as process assumptions as presented by Sánchez-Rivera et al. (2021), were adopted for this study to provide a basis for comparison. Previous research indicates that a solvent recovery rate of $\geq 99.9\%$ is necessary for the economic viability of ML plastic film separation via solvent-based processes (Sánchez-Rivera et al., 2021, 2023). However, the anti-solvent STRAP™ variant is characterized by lower recovery rates due to azeotrope formation during distillation. To avoid this challenge, the temperature-dependent STRAP™ variant was adopted in this study.

The detailed process flow diagram used for process simulation in this study is presented in Fig. 9. A description of this detailed process, the assumptions adopted, and detailed descriptions and functions of each process unit including the hot filtration systems, dryers, pumps, cooling tanks (precipitator), and condensers, are presented in the ESI. In summary, the process modeling, techno-economic analyses (TEA), and life-cycle assessments (LCA) of the solvent-based multilayer plastic separation process encompass three key stages: plastic dissolution, precipitation, and solvent recovery, as depicted in Fig. 10. Plastic dissolution is accounted for in the TEA and LCA analyses via the process equipment cost and utility requirements of the dissolution vessels. Precipitation is accounted for via the precipitation tanks and cooling utilities. Finally, solvent recovery is accounted for via the solvent storage tanks, filtration systems, dryers, condensers, and associated heating and cooling utilities. The detailed process conditions employed for each process (i.e., different solvent pairs) are also presented in Table 2.

The MSPs obtained from TEA are presented in Table 3. As a basis for comparison, using similar assumptions and process flow diagram, the result obtained from this study was in agreement with the work published by Sánchez-Rivera et al. (2021) as shown in Table 3. However, the slight difference is attributed to the different underlying assumptions between ASPEN and BioSTEAM simulators as well as the removal of the extrusion process in this study. The plastic film extrusion process was removed from this study because, at that stage,

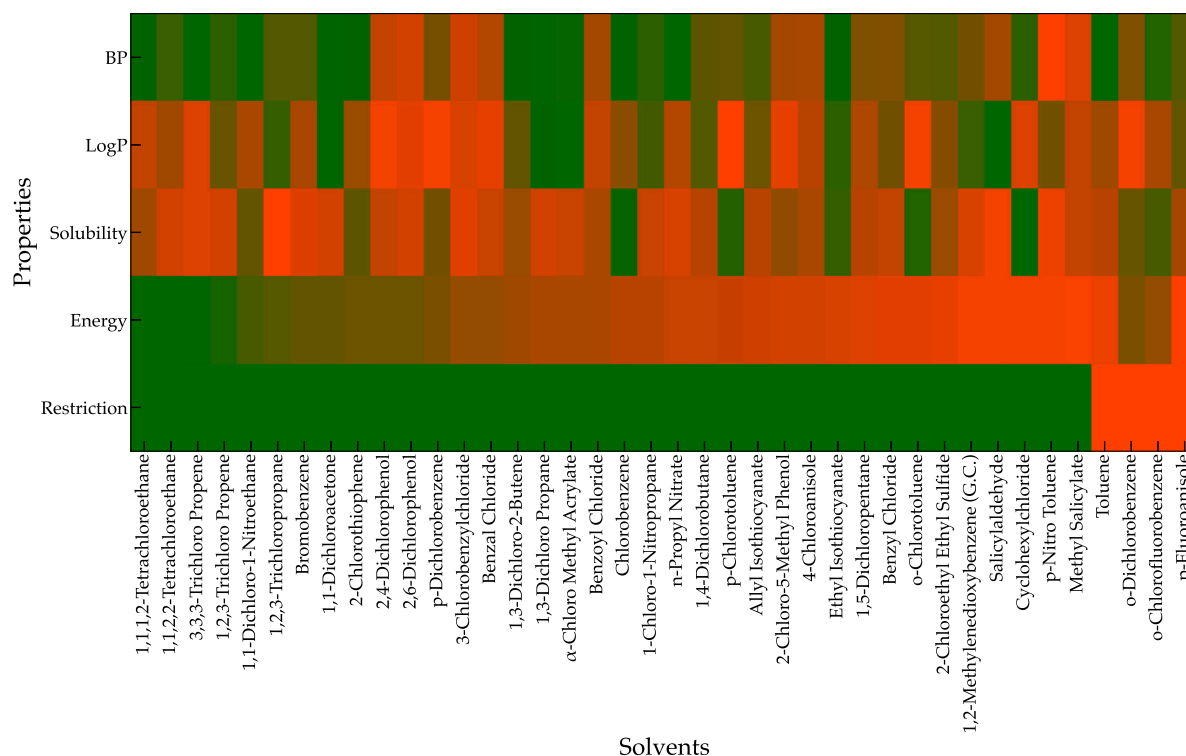


Fig. 8. Selected solvents for PE selective dissolution. The green color represents low boiling point, low logP, high solubility, low energy, and no restriction. BP: Boiling Point. The properties of each solvent are presented in the ESI.

Table 1

Selected solvent pairs for the separation of a PE-EVOH-PET ML plastic film and their rationale for selection. Potential solvents for the selective dissolution of PE (corresponding to Solvent A in Fig. 1) are shown in the PE (PE-EVOH-PET) column, and potential solvents for the selective dissolution of EVOH (corresponding to Solvent B in Fig. 1) are shown in the EVOH (EVOH-PET) column. S/N refers to serial number.

S/N	PE (PE-EVOH-PET)	EVOH (EVOH-PET)	Rationale
1	Toluene	DMSO	Published work (del Carmen Munguía-López et al., 2023; Sánchez-Rivera et al., 2021)
2	Tetramethylene sulfide	DMSO	Highest solubility ^a
3	Cyclopentanone	Acetic acid	Lowest logP ^a and best LCA metrics (Henderson et al., 2011)
4	o-Xylene	Epsilon-Caprolactam	Biomass-derived solvents (Hu et al., 2017; Buntara et al., 2011)
5	Cyclohexanol	Acetic acid	Best LCA metrics (GSK) (Henderson et al., 2011)
6	α-chloro methyl acrylate	Bromoacetic acid	Lowest energy requirement ^a
7	Toluene	DMF	Common solvents

^a Obtained from this study.

Table 2

Operating conditions used in each process to recycle 3000 tons/year ML plastic waste.

Process	Solvent A	Solvent B	Polymer/ solvent ratio	Dissolution Temp A (°C)	Dissolution Temp B (°C)	Dryer Temp A&B (°C)	Precipitation Temp A&B (°C)	Condenser Temp A (°C)	Condenser Temp B (°C)
Process I	Toluene	DMSO	20	110	95	100	25	−30	0
Process II	Tetramethylene sulfide	DMSO	20	120	95	100	25	−30	0
Process III	Cyclopentanone	Acetic acid	20	120	110	100	25	−30	−30
Process IV	o-Xylene	Epsilon-caprolactam	20	120	120	100	25	−30	60
Process V	Cyclohexanol	Acetic acid	20	120	116	100	25	0	−30
Process VI	α-Chloro Methyl Acrylate	Bromoacetic acid	20	120	120	100	25	−30	40
Process VII	Toluene	DMF	20	110	120	100	25	−30	−20

the solvent choice does not affect the additional cost since our goal is to compare the effect of solvent choice. Table 3 shows that MSPs do not significantly change with solvent pair choice, indicating the choice of solvent does not significantly impact the MSP when high solvent recovery rates are enforced. Details of the sensitivity analysis showing the effect of solvent recovery on process cost are presented in the ESI. This sensitivity analysis also showed that the solvent boiling point significantly affects the recovery rates of solvents. The detailed cost breakdown for each solvent choice is also presented in the ESI.

3.4. Life cycle assessment

Having selected potential solvents based on polymer recovery via dissolution and precipitation, energy demands, greenness via logP and regulatory guidelines, and the associated cost, as indicated by the framework introduced in this study, we present an LCA as another solvent greenness assessment. While other solvent greenness metrics are independent of their application within a process, an LCA offers the benefit of quantitatively analyzing solvent greenness within the context

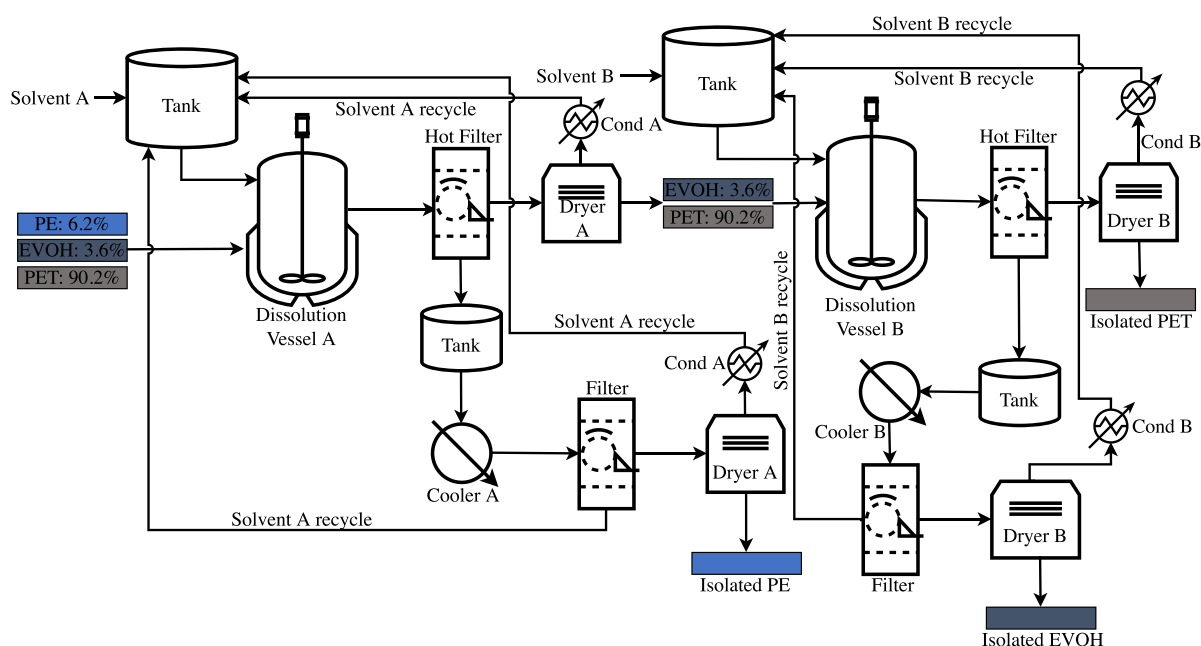


Fig. 9. Process flow diagram for PE-EVOH-PET ML plastic film separation via STRAP™. Cond: Condenser.

Table 3

Minimum selling price (MSP) for selected solvent pairs.

Process	PE (PE-EVOH-PET)	EVOH (EVOH-PET)	MSP (USD/kg)
0	Toluene	DMSO	1.62 ^a
I	Toluene	DMSO	1.44
II	Tetramethylene sulfide	DMSO	1.43
III	Cyclopentanone	Acetic acid	1.48
IV	o-Xylene	Epsilon-Caprolactam	1.45
V	Cyclohexanol	Acetic acid	1.46
VI	α -chloro methyl acrylate	Bromoacetic acid	1.34
VII	Toluene	DMF	1.48

^a Work by Sánchez-Rivera et al. (2021).

of the entire STRAP™ process. The LCA methodology principles adopted in this study are in compliance with the ISO-14040 and 14044-LCA series of standards (Finkbeiner et al., 2006). This LCA study quantifies the environmental benefits of using each solvent pair indicated in Table 1 within the STRAP™ process using climate change and process toxicity as metrics. As STRAP™ is seen as an alternative process for producing virgin-grade polymers, ensuring circularity (Walker et al., 2020; del Carmen Munguía-López et al., 2023), a product perspective is taken in this analysis (Nordahl and Scown, 2024), and the functional unit considered here is the production of 1 kg of virgin resins. Thus, we compare the impact of producing 1 kg of virgin resins using the different solvent pairs chosen. The system boundary for the LCA on the STRAP™ process is presented in Fig. 10. In this process, only the three polymers are assumed to be contained in the ML plastic film, as indicated by the film composition assumed in Fig. 1.

The impact categories in this study include climate change impact (kg CO₂ eq.), human toxicity cancer (CTUh), human toxicity non-cancer (CTUn), and freshwater ecotoxicity (CTUe). Climate change impact measures the global warming potential of greenhouse gases such as CO₂, NO_x, SO₂, N₂O, CH₄, and particulate matter. Human toxicity (cancer and non-cancer) indicates the disease rate increase in the human population due to exposure to chemicals. Freshwater ecotoxicity measures the impact of exposure to harmful chemicals on aquatic life in freshwater. As indicated in the LCA system boundary (Fig. 10), it is critical to note that this study does not include the impacts of producing

each of the solvents used in this case study. However, this approach can be adapted to this framework in future studies. The factors of the environmental impacts for each category were obtained using the environmental footprint (mid-point indicator) method from the OpenLCA 2.0 environmental footprint database. The necessary utility flow rates were obtained from the process simulation on BioSTEAM. All impact category factors were based on the European data to ensure fair comparison across the different processes (i.e., solvent pairs). This was due to the unavailability of virgin polymer resin impact factors based on the United States data.

The results from the LCA study for process I through process VII, as labeled in Table 2, are presented in Fig. 11. Fig. 11(a) shows the climate change impacts (CCI) for each process employing different solvent pairs. The average CCI for the different processes using various solvents was observed to be 1.0 kg CO₂ eq./kg of ML film. This value differs slightly from the CCI of 0.5 kg CO₂ eq./kg of ML film reported by del Carmen Munguía-López et al. (2023) for a similar ML film separation process using the same STRAP™ process variant. This discrepancy is primarily due to differences in the databases used for impact factor calculations and the use of data obtained from a different process simulator (Aspen), despite both studies reporting a similar process energy use of 10 MJ/kg of film. It was also observed that Process VI, using α -chloro methyl acrylate and bromoacetic acid, achieved the lowest CCI of 0.8 kg CO₂ eq./kg of film. This solvent pair was originally chosen because they had the lowest solvent energy requirement. This observation hints at the fact that the solvent energy requirement, estimated from its heat capacity, is a major driver of the CCI of a process, except if the energy is sourced from renewables (del Carmen Munguía-López et al., 2023).

The human toxicity (cancer/non-cancer) results are presented in Figs. 11(b) and 11(c). A similar trend was also observed with Process VI recording the lowest human toxicity cancer and non-cancer of 4.06×10^{-10} and 8.44×10^{-9} CTUh/kg of resin, respectively. Similar results with the same order of magnitude were reported in the literature (del Carmen Munguía-López et al., 2023). The freshwater ecotoxicity followed the same trend as shown in Fig. 11(d). In comparison to the impacts of virgin film production from fossil fuel

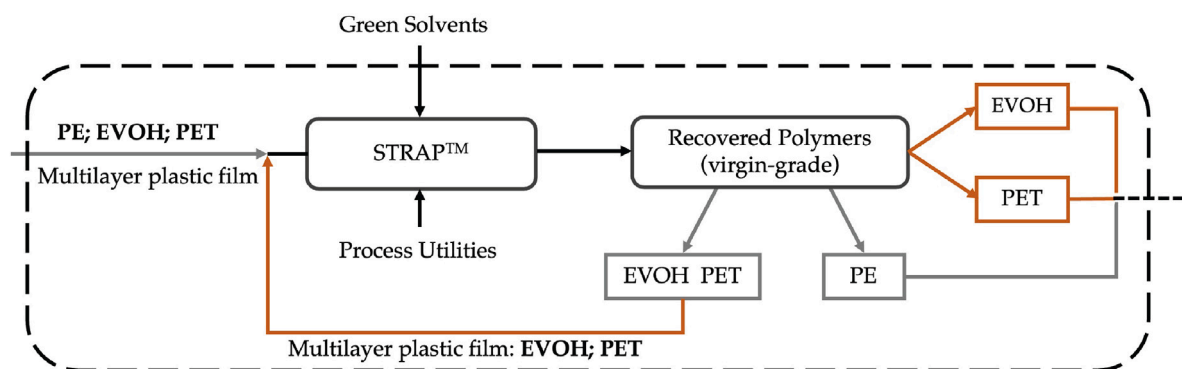


Fig. 10. System boundary for the LCA study on the STRAP™ process. The gray and orange lines differentiate the first and second recovery steps.

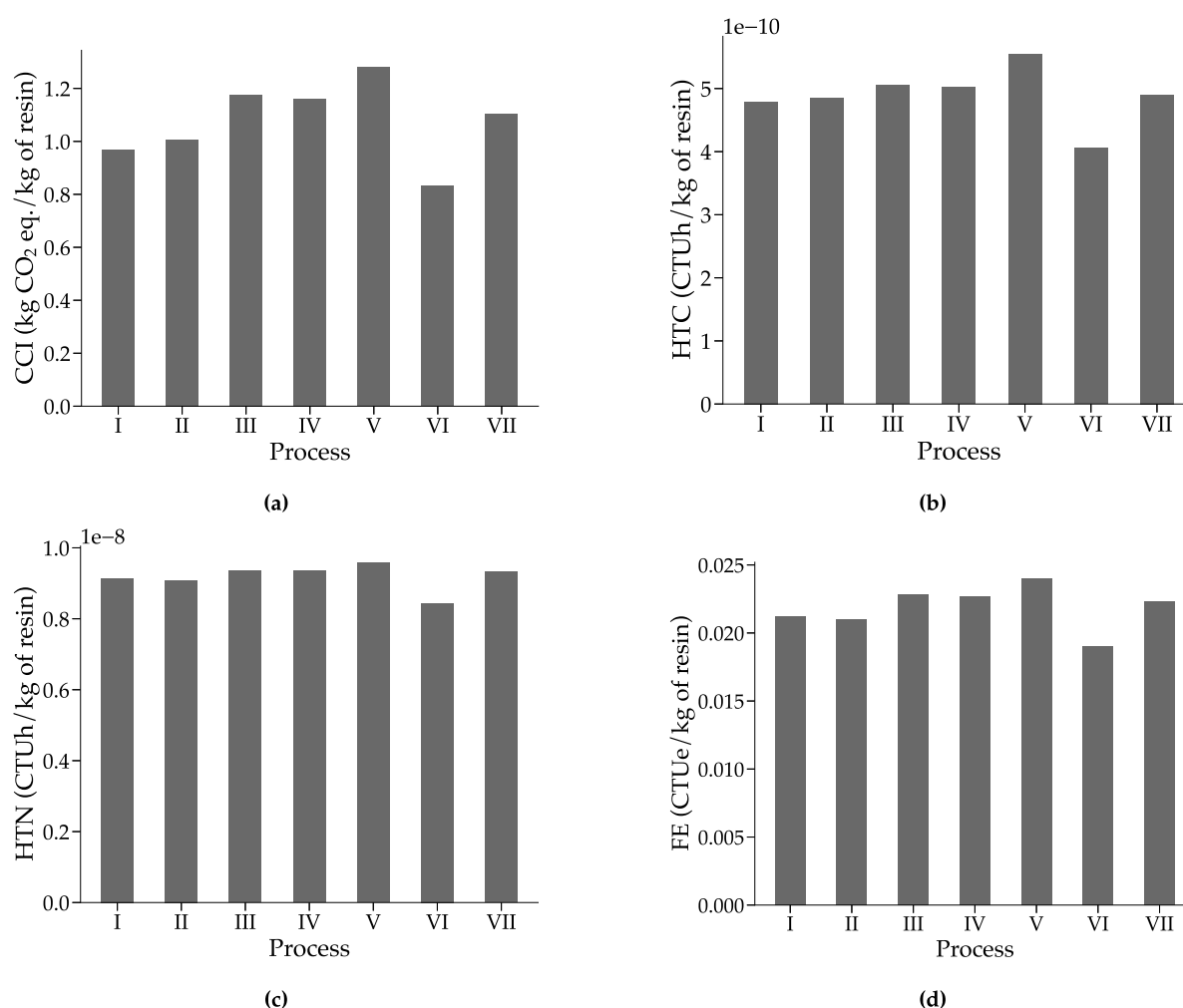


Fig. 11. (a) CCI: Climate change impacts, (b) HTC: human toxicity cancer, (c) HTN: human toxicity non-cancer, and (d) FE: freshwater ecotoxicity. See Table 3 for solvent pairs used in each process.

as reported by del Carmen Munguía-López et al. (2023), our study supports the claim that STRAP™ offers an environmentally friendly alternative for producing polymer resins. For instance, our study shows a 54% reduction in climate change impact. In addition to the impact factors of the utilities considered in this study, the impacts resulting from the production of each solvent would significantly affect the results. For instance, solvents that are considered biomass-based could have either a net-zero or negative CCI, thereby reducing the over CCI for that process, depending on some factors such as the biomass source, etc.

3.5. Trade-off analysis between solvent greenness metrics

Having applied all the steps of our framework, we consolidated all the obtained solvent greenness metric results and performed a trade-off analysis comparing the seven processes, intending to identify the best green solvent pair for ML plastic film separation via STRAP™. Fig. 12 displays the solvent greenness metrics for each of the seven processes using different solvents, presented on a radar chart. The metrics include the MSP of the recovered virgin resins, the average logP and the combined energy requirement of both solvents in the pair, their

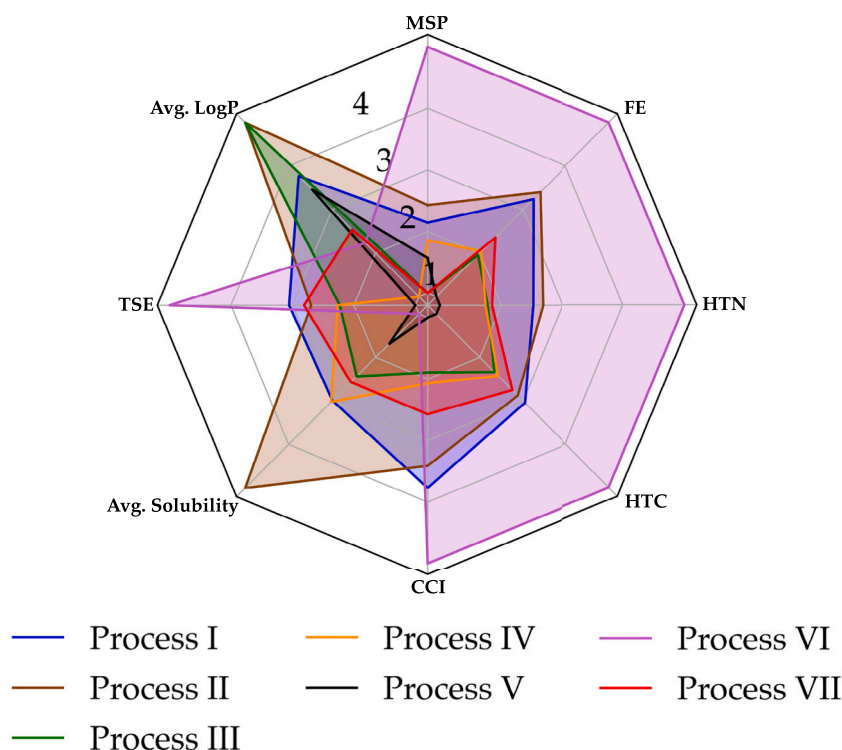


Fig. 12. Comparative analysis of seven processes employing different solvent pairs (listed in Table 3) using eight solvent greenness metrics. MSP: Minimum Selling Price, Avg. logP: Average Octanol-Water Partition coefficient, TSE: Total Solvent Energy, CCI: Climate Change Impact, HTC: Human Toxicity (Cancer), HTN: Human Toxicity (Non-cancer), FE: Freshwater ecotoxicity.

average solubility, climate change impact, human toxicity (both cancer and non-cancer), and freshwater ecotoxicity. While other metrics like the individual solvent's boiling point are important, using the average logP, solubility, and total solvent energy requirements provides a more intuitive comparison when dealing with solvent pairs rather than an average boiling point.

The actual values of these estimated metrics are presented in the ESI. Each metric for all solvent-pairs is normalized between 1 and 5 for comparison and is plotted on the radar plot. A score of 5 indicates the most green solvent pair, while 1 indicates the least green solvent pair. In summary, we seek to find a solvent-pair that covers the most area on the radar plot. As shown in Fig. 12, no solvent pair, and thus no solvent, performs perfectly in all greenness metrics. Process VI (α -chloro methyl acrylate & bromoacetic acid) ranks best in MSP, FE, HTN, HTC, CCI, and TSE metrics. However, it has the lowest average solubility (17.33 wt.%) and a relatively high (positive) average logP (0.90). This compromise, necessary to meet other greenness metrics, is referred to as the “cost of greenness”. When we compare Table 1 with Fig. 12, we see that prioritizing solubility, Process II (Tetramethylene sulfide & DMSO), also ranks best under the logP metric, however, it performs lower than Process I (Toluene & DMSO) and Process VI under some other solvent greenness metrics.

Furthermore, we see that although the GSK solvent guideline identifies cyclohexanol and acetic acid (Process V) as having the best LCA metrics based on their study (Henderson et al., 2011), our study finds these to be the least green solvents compared to the other evaluated solvents (considering climate change impact, human toxicity and ecotoxicity as metrics). It is important to note that the GSK life cycle score evaluates the environmental impacts of solvent production and their environmental fate and effects. In contrast, our study assesses the life cycle and environmental impacts of using solvents in an ML plastic film separation via a solvent-based process. This discrepancy underscores the importance of considering both production and usage impacts within our frameworks when relevant data are available. Fig. 12 also reveals the relationship between the total solvent energy

required and the LCA impact categories. We observe that the TSE and the LCA impact categories are highly correlated. This observation agrees with the correlation analysis on all eight solvent greenness metrics presented in the ESI. This suggests that designing solvents with lower heat capacities that exhibit high selective solubility at mild temperatures should be considered to improve the LCA impacts. Also, prioritizing logP (i.e., Process III), a toxicity measurement, did not result in a good score on other metrics. This observation indicates that the logP of solvents does not correlate with other solvent greenness metrics, as shown in the correlation matrix in the ESI.

Based on the results obtained in this study, solvents used in Processes I, II, and VI exhibit the best greenness metrics. However, it is crucial to qualitatively assess the hazards of these solvents. For example, bromoacetic acid is corrosive, acutely toxic, and exhibits serious environmental hazards. Similarly, α -chloro methyl acrylate is flammable, acutely toxic, and corrosive. Toluene presents significant health risks and is unsuitable for recovering food contact materials; it is also listed as a restricted substance under the REACH initiative. Both tetramethylene sulfide and DMSO are only classified as irritants. Therefore, Process II offers the least qualitative hazards and a high greenness ranking. Additionally, the bio-based solvents used in Process IV (o-xylene and epsilon-caprolactam) may benefit the LCA impact categories. Similarly, Cyrene™, an aprotic dipolar solvent derived from waste biomass (Camp et al., 2020), has been recommended as a green alternative to DMSO with similar functionalities. However, the selective solubility of EVOH in Cyrene™ is yet to be determined. In conclusion, this framework effectively identifies green solvents and highlights trade-offs between solvent greenness metrics in ML plastic film separation via a solvent-based process.

3.6. Application of computational framework for process design

The design of a process often takes precedence before an LCA and TEA are carried out to evaluate and improve the environmental

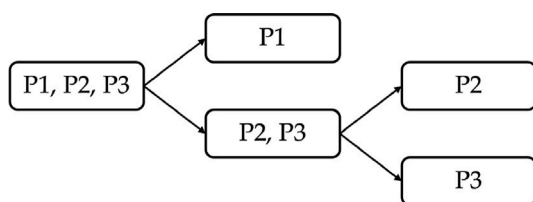


Fig. 13. Separation sequences for an example 3-component multilayer plastic film. P1, P2, P3 each correspond to one of PE, EVOH, or PET.

Table 4

Screened solvents for different multilayer plastic film separation process designs.

Sequence	No. of available solvents	
	Step 1	Step 2
PE-EVOH-PET	151	45
PET-EVOH-PE	0	–
EVOH-PE-PET	21	171
PE-PET-EVOH	151	4
PET-PE-EVOH	0	–
EVOH-PET-PE	21	1

impact and feasibility of the process. In this section, we illustrate the use of our framework to guide the design of a sustainable process for ML plastic film separation using the same STRAP™ case study by assessing green solvents and possible separation sequences. This approach sought to integrate sustainability considerations from the outset to guide the process design. Also, having identified the trade-offs from the case studies considered in the previous sections, we observe that selecting a green solvent for ML plastic film separation requires a multiobjective approach to avoid burden-shifting (Algunaibet and Guillén-Gosálbez, 2019). Therefore, designing a process to minimize the risk of burden-shifting in ML film separation is required from the outset.

To achieve a more sustainable process design for the case study considered in this work, our framework will be applied to the six possible sequences for separating the PE-EVOH-PET ML plastic film. It is critical to note that this study only sought to select the most sustainable separation sequence without optimizing the energy requirements or design of unit operations in the detailed process design. As we can observe, these possible sequences can quickly scale up as the number of polymer components (n) increases by n -factorial. Fig. 13 shows the separation sequences that can be adopted for a 3-component ML plastic film separation. “P” corresponds to the polymer components, and the six different separation sequences are PE-EVOH-PET, PET-EVOH-PE, EVOH-PE-PET, PE-PET-EVOH, PET-PE-EVOH, and EVOH-PET-PE. Prior to the development of this framework, previous studies adopted the PE-EVOH-PET sequence as depicted in Fig. 1 because of its ease of operation. However, considering the ease of operation as well as the environmental impact of the different separation designs will be beneficial.

The solvents to achieve each separation process design were quickly screened following the quantitative section of our framework. After screening the solvents based on selective dissolution, precipitation, and the logP criteria as described in Fig. 2, Table 4 presents the number of solvents that can achieve each separation step for the different separation process designs. Step 1 corresponds to the selective dissolution of the first component, and step 2 is the selective dissolution of the second component, leaving the third component undissolved.

PE-EVOH-PET sequence corresponds to the separation process design studied in the previous section. Based on our previous findings, 151 solvents were identified to achieve the step 1 dissolution process, while 45 solvents were identified to achieve the step 2 dissolution process. Since this design has been studied in the previous sections, the best solvent pair (tetramethylene sulfide and DMSO) will be compared

Table 5

Selected solvent pairs for different sequences.

Sequence	Solvents	
	Step 1	Step 2
PE-EVOH-PET	Tetramethylene sulfide	DMSO
EVOH-PE-PET	DMSO	Tetramethylene sulfide
PE-PET-EVOH	Tetramethylene sulfide	Ethyl thiocyanate
EVOH-PET-PE	DMSO	Pyrrole

with other identified solvent pairs capable of achieving the other designs. It was observed that no solvent within our database was able to achieve PET-EVOH-PE and PET-PE-EVOH separation process design, as shown in Table 4. This is due to the difficulty associated with the selective dissolution of PET (Sherwood, 2020; Suhaimi et al., 2022). Therefore, based on the selective dissolution, precipitation, and the logP criteria, only PE-EVOH-PET, EVOH-PE-PET, PE-PET-EVOH, and EVOH-PET-PE designs are feasible, reducing the number of possible separation designs.

Secondly, the selected solvents were further screened following the qualitative section of our framework. This screening includes assessing solvents based on their qualitative hazards. For instance, about half the solvents identified to achieve EVOH-PE-PET design are acids, making them unfit for recovering food-grade materials. Therefore, this section aimed to select solvent pairs with the least hazards for each sequence to conduct trade-off analyses using the eight solvent greenness metrics introduced in Fig. 12.

Table 5 presents the selected solvent pairs. First, we observe that the solvent space for the selective dissolution of PE is relatively large. We also observe that similar solvents in the reverse order can achieve both PE-EVOH-PET and EVOH-PE-PET designs. This is because tetramethylene sulfide is selective to PE, while DMSO is selective to EVOH. Furthermore, we also observe that while four solvents can selectively dissolve PET from a 2-component system, only one solvent is worth evaluating. Other solvents include 1,2-ethane dithiol, ethyl isothiocyanate, and methyl isothiocyanate, which possess severe environmental hazards, health hazards, corrosive nature, and are acutely toxic. This qualitative hazard further reduces the solvent space.

Lastly, similar to the case study trade-off analysis, a comparative analysis is done on the down-selected sequences using the solvent pairs presented in Table 5. This involved performing a TEA and LCA on each process design. As in the case studies, the polymer composition was kept the same when performing these analyses and all other process design assumptions were kept the same. However, each process design was adapted to match the sequence in which the ML plastic film is separated. For instance, the PE-EVOH-PET design involves first selectively dissolving PE followed by EVOH and finally recovering PET. In contrast, the EVOH-PET-PE design involves first selectively EVOH followed by PET and finally recovering PE.

The trade-offs of the four possible process designs are presented in Fig. 14. The actual values for each solvent greenness metric are presented in the ESI. Similar to Fig. 12, 1 represents the least green sequence for each metric, while 5 corresponds to the most green. We observed that the PE-EVOH-PET and EVOH-PE-PET designs yielded the same results after performing the TEA and LCA. This is shown as both lines overlap in Fig. 14. This observation suggests that as long as the polymer composition of an n -component ML plastic film remains constant, the order of recovering the $n-1$ components does not impact the economics or environmental effects, provided that the same final polymer remains undissolved.

Furthermore, it was also observed that while the total solvent energy (TSE) for the EVOH-PET-PE design was the highest (389.80 J/g), it did not perform the worst under the climate change impact (CCI) and the freshwater ecotoxicity (FE) metrics. Similarly, although the PE-PET-EVOH design had the lowest TSE with 348.80 J/g, it did not perform the best under the CCI and HTN metrics. These observations show

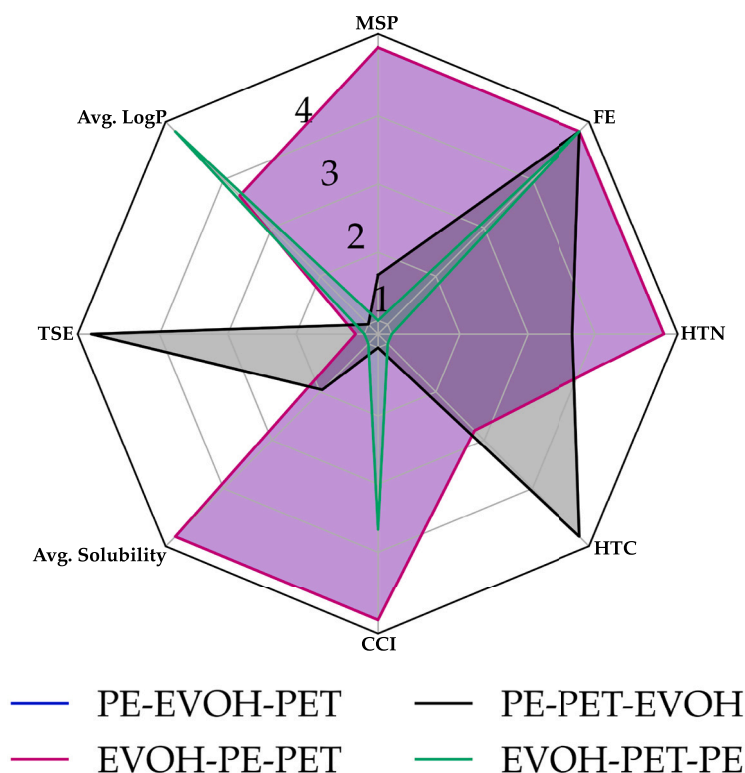


Fig. 14. Comparative analysis of four possible process designs employing different solvent pairs using eight solvent greenness metrics. PE-EVOH-PET and EVOH-PE-PET designs overlap with each other. MSP: Minimum Selling Price, Avg. logP: Average Octanol-Water Partition coefficient, TSE: Total Solvent Energy, CCI: Climate Change Impact, HTC: Human Toxicity (Cancer), HTN: Human Toxicity (Non-cancer), FE: Freshwater ecotoxicity.

that although energy correlates with these LCA metrics, as observed previously in this study, this trend is not necessarily observed when the difference in the TSE of solvents in the processes is not significant. In conclusion, PE-EVOH-PET and EVOH-PE-PET outperform other process designs. However, the ease of operation from an experimental standpoint can be further applied to compare both process designs.

Having highlighted the advantages of organic solvents in selectively targeting polymer components in multilayer plastics for effective separation, this study identifies their significant health and environmental concerns. Some key properties for an optimal solvent include high selective solubility of target polymers at mild temperatures, low energy requirements, a high boiling point (except in the case of solvent mixtures), negative octanol-water partition coefficients (LogP), and biomass-derived origins. However, applying this framework could potentially lead to not finding solvents with satisfactory performance. Hence, computer-aided molecular design (CAMD) can be adapted to expand this framework. This can be achieved via two approaches, namely: computer-aided solvent design and computer-aided multilayer plastic design.

The first approach involves the use of traditional physics-based models and CAMD approaches to design and identify better solvents for specific applications. For instance, [Struebing et al. \(2013\)](#) combine quantum mechanics (QM) studies with CAMD approaches to design solvents to accelerate the reaction kinetics and study solvents effects in reactions. More recently, model-based design of experiments was integrated with a QM-CAMD framework to reduce the computational cost associated with designing and identifying solvents via traditional QM methods ([Gui et al., 2023](#)). CAMD methods have also found applications in designing solvent mixtures for dissolving additives in paints ([Gani, 2007](#)). Molecular-scale studies have also been combined with data-centric models to quickly design and predict glycerol-based solvents that can be synthesized from biomass and deep-eutectic solvents with favorable properties for polymer dissolution ([Soyemi and](#)

[Szilvási, 2023; Christodoulou et al., 2024](#)). Given the preferred properties of a green solvent for multilayer plastic separation as identified by our framework, these tools can be adopted to design solvents with satisfactory performance.

Given the challenge of separating PET in recycling processes as highlighted in this study, the second approach involves integrating molecular-scale polymer studies with data-driven predictions to inform the optimal design of easily recyclable ML plastics that is fit for purpose. For instance, [Munguía-López et al. \(2024\)](#) developed a computational framework to quickly identify problematic ML plastic compositions and propose alternatives that reduce environmental impact and cost. The optimal design of recyclable polymers to fit specific applications have been explored in the literature. For instance, [Rorrer et al. \(2019\)](#) studied the combination of PET waste with bio-based monomers and observed both an improvement in its properties and its recyclability. Similarly, [Shi et al. \(2021\)](#) discussed in detail, the principles for designing next-generation intrinsically circular polymers with properties tunable for different applications. This design strategy follows the plastic redesign concept called *recyclable-by-design (RBD)*, ensuring easy recyclability of these plastics at their end-of-life. This approach enables the redesign of polymers compatible with green solvents, reducing reliance on screening organic solvents to dissolve legacy polymers.

4. Conclusions and future work

This study presents a green solvent selection framework for multilayer (ML) plastic film separation via solvent-based processes. The framework addresses potential environmental risks, focusing on mitigating new toxic waste generation. A case study on PE-EVOH-PET ML plastic film separation using a temperature-dependent STRAP™ process demonstrates the framework's effectiveness.

The eight-step framework begins by predicting temperature-dependent polymer solubilities in 1000 solvents. Solvents are screened based on their ability to selectively dissolve and precipitate individual

polymers at ambient pressure. Solvent greenness is assessed using both quantitative and qualitative methods. Quantitatively, the solvents' logP and GSK solvent guide scores were used for rapid green solvent screening, while environmental impacts, including climate change, human toxicity (cancer & non-cancer), and freshwater ecotoxicity, were evaluated via LCA. Qualitatively, solvent hazards were reviewed from public databases and regulatory frameworks such as REACH and EPA. Process costs associated with each solvent choice are evaluated and assessed via a TEA. This framework was applied to a 3-component ML plastic film, resulting in two separation processes: selective dissolution of EVOH from EVOH-PET and PE from PE-EVOH-PET. The framework also guided process design by evaluating all possible separation sequence designs.

For the binary ML system involving selective dissolution of EVOH from an EVOH-PET film, we identified 45 out of 1000 solvents that met both selective dissolution and logP criteria. DMSO, previously used in the STRAP™ process, outperformed most solvents in terms of energy requirement, solubility, and logP. Although formic acid showed lower energy requirements and logP, its corrosiveness makes it less suitable for recovering food contact materials. In the 3-component system involving selective dissolution of PE from a PE-EVOH-PET polymer mix, 151 out of 1000 solvents met the selective dissolution and logP criteria. Toluene, despite being flagged as a chemical of concern in the GSK solvent guidelines, also performed poorly in terms of energy requirements and logP. Solvent boiling points were found to significantly affect recovery rates, influencing the economics and sustainability of solvent-based processes. High boiling point solvents should be avoided due to energy-intensive separation, while low boiling point solvents may hinder recovery rates. At high solvent recovery rates of $\geq 99.90\%$, the choice of solvent had a negligible effect on the MSP of recovered resins. However, below 99.90% recovery, the solvent contribution to the variable cost became significant. The LCA metrics indicated that a solvent's minimum energy requirement, which is a function of its specific heat capacity, is strongly correlated with environmental impact. A trade-off analysis showed that, while no significant improvements were observed in MSP and LCA metrics, tetramethylene sulfide emerged as a potential replacement for toluene based on qualitative hazard assessment. We further demonstrated how this framework can support more sustainable process design for ML plastic film separation by evaluating different separation sequences. PET presented the greatest challenge in selective dissolution, remaining undissolved at the final separation stage. Screening for green solvents further reduced the solvent space for separation, highlighting the complexity of selecting a green solvent for ML plastic film separation as a multiobjective problem.

Future work could investigate methods to combine our solubility database, this framework, and BioSTEAM, a process simulator, to rapidly screen for green solvents for separating ML plastic films with more polymer components while also considering potential contaminants including the intentionally and non-intentionally added substance (Tumu et al., 2024). Although this study applies our green solvent selection framework to an ML plastic comprising of three polymers, this framework can be applied to ML plastics comprising of up to 10 polymers studied by Sánchez-Rivera et al. (2025) as their temperature-dependent solubilities currently exist in our database. However, in the case where these temperature-dependent solubilities do not exist, our database can be extended to more polymers, employing similar physics-based solubility prediction workflow described by Zhou et al. (2023). One limitation encountered in this study was the insufficient quantitative approach to rapidly screen green solvents. There are potential opportunities to employ data science techniques to expand the GSK solvent screening guide to the 1000 solvents in our database. These quantitative metrics will aid the development and solving of multiobjective optimization models for green solvent selection in solvent-based processes. As mentioned earlier, this study employs a temperature-dependent solvent-based ML plastic separation process, avoiding the cost and environmental impacts associated with

separating solvents via a distillation process and azeotrope formation. However, in the case where anti-solvents are employed, we recommend employing computational methods to quickly predict azeotrope mixtures and implementing a constraint to avoid such solvent combinations during the solvent selection process. As part of future work, we also are interested in using this framework to identify easy-to-recycle ML plastic films from a technical and sustainability standpoint. This will guide the design of more sustainable products and food packaging materials.

CRediT authorship contribution statement

Ugochukwu M. Ikegwu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Aurora del Carmen Munguía-López:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Victor M. Zavala:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Reid C. Van Lehn:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compchemeng.2025.109129>.

Data availability

Data will be made available on request.

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